

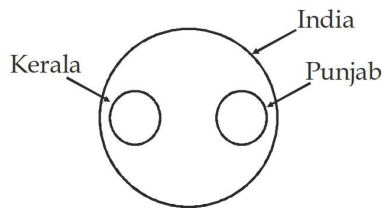
Logical Venn-Diagram

A Venn diagram is a diagram that shows all possible logical relation between a finite collection of different sets of objects through a certain diagram. Relations between the sets can be indicated by the arrangement of certain diagrams. In short, it is a diagram that uses circles to represent sets and their relationship

In this topic, two types of questions are generally asked;

(a) Venn diagram of sets of different objects.

Ex.:- India, Kerala, Punjab.

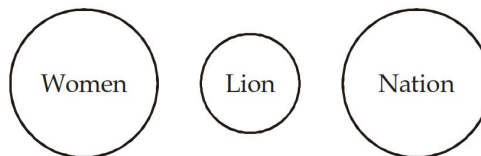


(b) Giving answers of the question from the given mixed diagram of two or more figures.

There are some basic types of Venn diagram that shows different relationships.

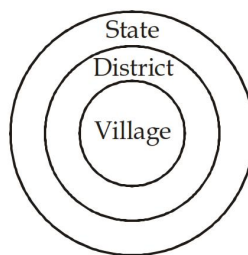
1. If all the items given in the question belong to different group then it will be shown by diagram as given below.

Ex. :- Women, Lion, Nation



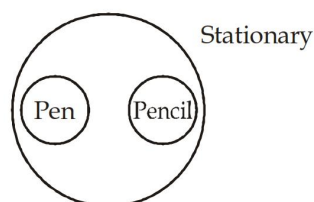
2. If the first item is related to the second item and second item is related to the third item, then all three items will be shown by diagram as shown below :-

Ex. :- Village, District, State.



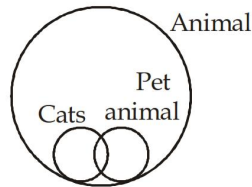
3. If two different items are related to third item, then they will be shown by diagram as given below.

Ex. :- Pen, Pencil, Stationary.



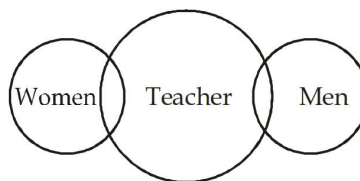
4. If there is some relationship between two items and these two items are completely related to the third items, then they will be shown by diagram as given below.

Ex. :- Animal, Pet animal, Cats



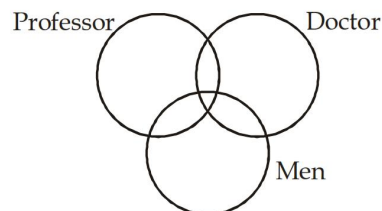
5. If two items are completely different from each other but are related to some extent but not completely to a third item, then they will be shown by diagram as given below.

Ex. :- Teacher, Men, Women



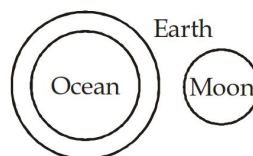
6. If all three items are related to each other to some extent but not completely, then they will be shown by a diagram as shown below:

Ex:-Men, Professor, Doctor.



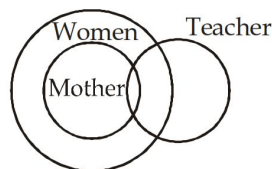
7. If the two items are completely related to each other and third item is entirely different from the other two items, then they will be shown by diagram as given below:

Ex:- Ocean, Earth, Moon.



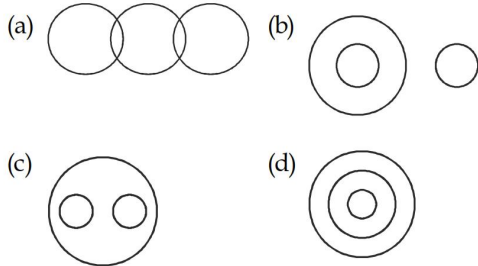
8. If the first item is related to the second item completely and the third is related to first and second item to some extent, then they will be shown by diagram as given below:

Ex:-Women, Teacher, Mother.



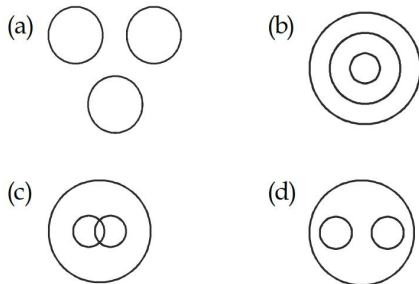
Solved Examples

1. Which one of the following diagram best depicts the relationship among planets, Jupiter and Moon?



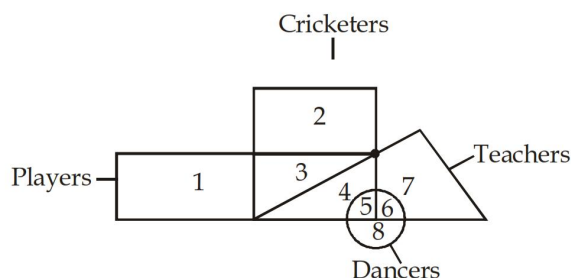
Sol. (b); Planet is category and Jupiter is a planet. We can put Jupiter in planet category. Moon is not a planet so we can put it outside the planets category

2. Which diagram represents the relationship among flowers, lotus and Lily ?



Sol. (d); Flower is a common noun and also a category itself. Lotus cannot be lily and lily cannot be lotus. So, we consider flower as a category and put lotus and lily differently.

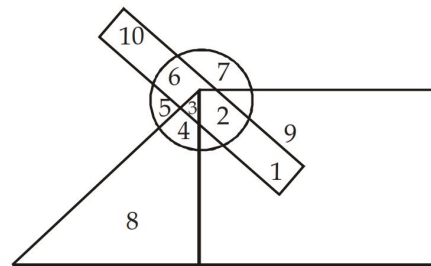
3. In the given diagram, rectangle represents players, square represents cricketers, triangle represents teachers and circle represents dancers, then which number represents the players, cricketers, teachers and dancers?



- (a) 2 (b) 5
(c) 6 (d) 3

Sol. (b); Number 5 represents all.

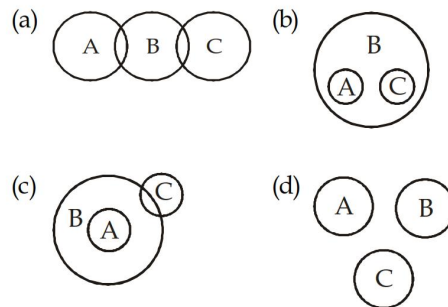
4. In the following figure, the boys who are athletes and disciplined are indicated by which number? The triangle represents girls, circle athletes, the rectangle boys and the square disciplined.



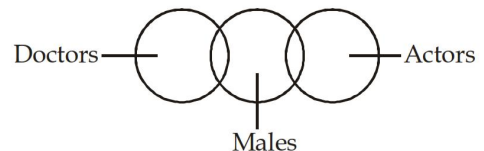
- (a) 1 (b) 2
(c) 6 (d) 10

Sol. (b); 2 represent the boys who are athletes and disciplined.

5. Which of the following figure/ diagram best depicts the relationship among doctors, Males, Actors?



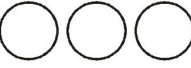
Sol. : (a);

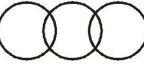


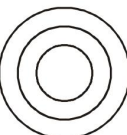
Some doctor and some actors are males. But, doctors and actors are entirely different.

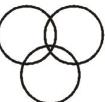
Practice Set

1. Which one of the following diagrams best depicts the relationship among teachers, graduates and women ?

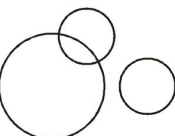
(a) 

(b) 

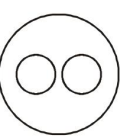
(c) 

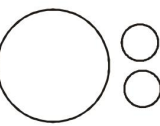
(d) 

2. Which of the following diagrams best depicts the relationship between house, bedroom and bathroom. (The sizes of circles do not indicate relative size of above)

(a) 

(b) 

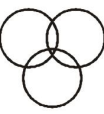
(c) 

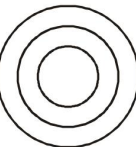
(d) 

3. Which one of the following diagrams best depicts the relationship among pigeons, birds and dogs ?

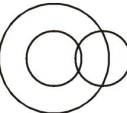
(a) 

(b) 

(c) 

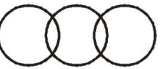
(d) 

4. Which one of the following figures represents the relationship among Males, Fathers, Advocates?

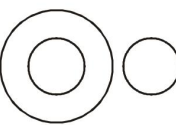
(a) 

(b) 

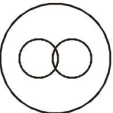
(c) 

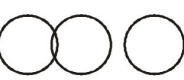
(d) 

5. Which one of the following figures represents the relationship among Dog, Animal, Snake ?

(a) 

(b) 

(c) 

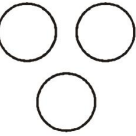
(d) 

6. Which of the following figure indicates relation between Aeroplane, Train, Travel ?

(a) 

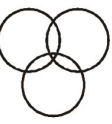
(b) 

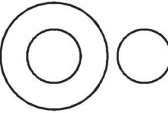
(c) 

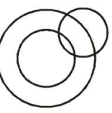
(d) 

7. Which of the following diagram best depicts the relationship between authors, teacher, men.

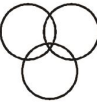
(a) 

(b) 

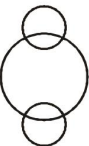
(c) 

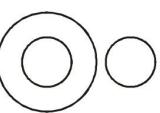
(d) 

8. For the given set of elements : Keyboard, Function keys, letter-keys.
Which figure given below will best represent the relationship among these three elements ?

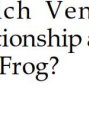
(a) 

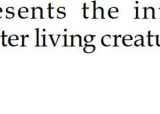
(b) 

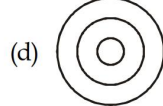
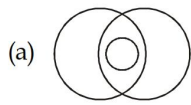
(c) 

(d) 

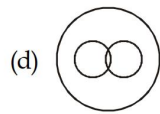
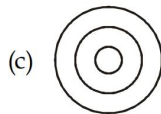
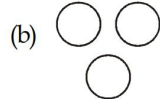
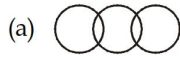
9. Which Venn diagram represents the inter-relationship among Reptiles, Water living creatures and Frog?

(a) 

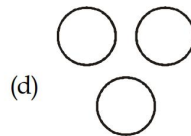
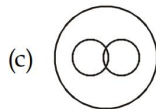
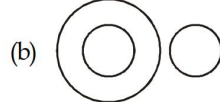
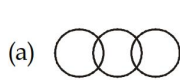
(b) 



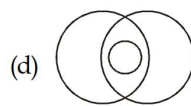
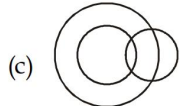
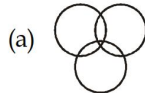
10. Which of the following diagrams best depicts the relationship between Book, Pen and Pencil?



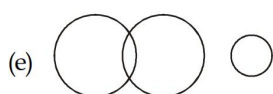
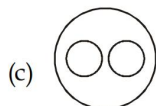
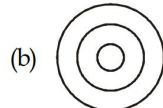
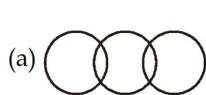
11. Which one of following diagram best depicts the relationship among Mountain, Earth, Forest?



12. Which of the following diagram represents the relationship between Poets, Dancer and Singers?



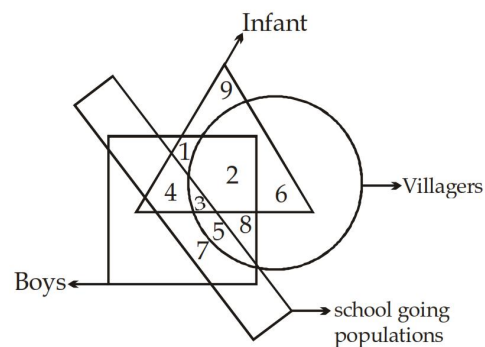
Direction (13 – 22): In each of these questions, three words are related in some way or other. The relationship among the words in question can be best represented by one of the five diagram (1), (2), (3), (4) and (5) given below. Choose that correct diagram.



13. Musician, Instrumentalists, Violinists
 14. Smokers, Lawyers, Non-smokers.
 15. Star, Earth, Sun
 16. Cricket player, Football player, Non-Football player
 17. Poet, Hockey player, Tennis ball
 18. Insects, Fly, mosquito
 19. Iron, Metal, Chlorine
 20. Ailments, Anorexia, Cataract
 21. Astrologer, Actor, Agriculture
 22. Relative, Son, Shirting

Direction (23 – 27): Read the following information carefully and answer the questions based on them.

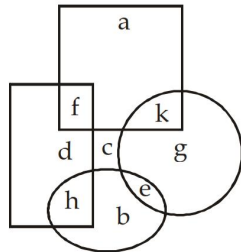
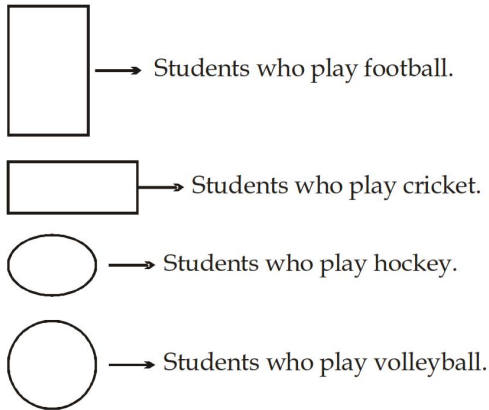
In the Venn-diagram given below the triangle represents infants, the square represents Boys, the circle represents villagers and the rectangle represents school going populations. Your task is to study the diagram and answer the questions that follow.



23. Which number represents non-school going villagers infants ?
 (a) 1 (b) 2
 (c) 6 (d) 2,6
24. Which number represents only villager boys?
 (a) 8 (b) 2
 (c) 1,2 (d) 2,6
25. Which number represents the school-going villagers boys ?
 (a) 3 (b) 5
 (c) 3,4 (d) 3, 4, 5, 7
26. Number 4 represents –
 (a) School going boys
 (b) Infants who are boys
 (c) Infants who are not villagers
 (d) School going boys who are not villagers
 (e) None of these
27. Which number represents the infant villagers boys who are school going?

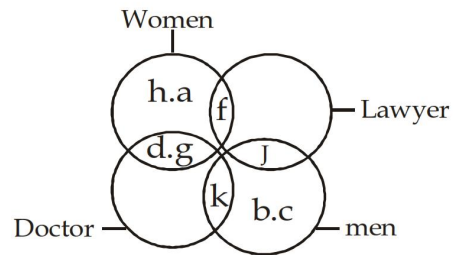
- (a) 2 (b) 3
(c) 5 (d) 7

Direction (28 – 32): Read the following information carefully and answer the question based on it.

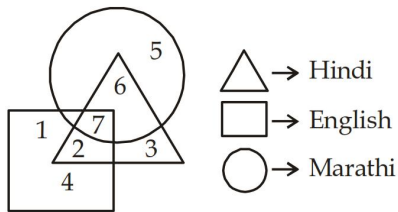


28. Letter c presents ?
(a) All students who play all matches.
(b) Students who play volleyball.
(c) Students who play cricket.
(d) None of these.
29. Which letter represents the students who play volleyball and hockey?
(a) e (b) h
(c) l (d) f
30. Which letter represents those students who play all games like volleyball, football, cricket and hockey?
(a) h (b) d
(c) a (d) None
31. Which letter represents the students who play football and cricket?
(a) e (b) a
(c) f (d) g
32. Which letter represents the student who play hockey and football?
(a) a (b) e
(c) f (d) None

Direction (33 – 37): Study the diagram given below to answer the questions.

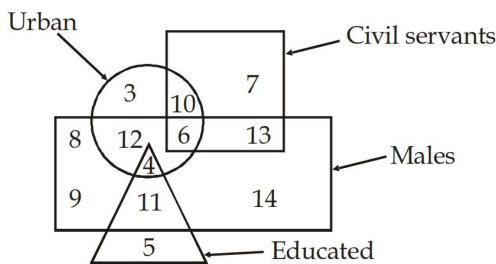


33. Which letter represents the lawyer women ?
(a) g (b) h and a
(c) f (d) None
34. Which letter represents the men doctor ?
(a) b (b) k
(c) b and c (d) c and a
35. Which letter represents such women who are not lawyer and doctor ?
(a) h, a (b) d
(c) f (d) q
36. Which letter represents the men who are not doctor and lawyer?
(a) j and k (b) h and a
(c) b and c (d) b and k
37. Which letter represent the women doctor ?
(a) h and a (b) d and g
(c) h and j (d) f
38. Which of the following diagram indicates the best relation between, Elephant, Carnivorous and Tiger.
- (a) (b)
(c) (d)
39. Which one diagram shows best relation among students, boys and girls.
- (a) (b)
(c) (d)
40. In the following diagrams the boys who are disciplined students.



- (a) 1 (b) 7
(c) 6 (d) 10

Direction (41- 48): The following questions are based on the diagram given below :-



- Rectangle represents males
- Triangle represents educated
- Circle represents urban
- Square represents civil servants

41. Who among the following is an educated male who is not an urban resident ?
(a) 4 (b) 5
(c) 9 (d) 11
42. Who among the following is neither a civil servant nor educated but is urban and not a male?
(a) 2 (b) 3
(c) 6 (d) 10
43. Who among the following is a female, urban resident and also a civil servant ?
(a) 6 (b) 7
(c) 10 (d) 13
44. Who among the following is an educated male who hails from urban area ?
(a) 4 (b) 2
(c) 11 (d) 5
45. Who among the following is an uneducated and also an urban male ?
(a) 2 (b) 3
(c) 11 (d) 12
46. Who among the following is only a civil servant but not a male nor urban oriented and uneducated?
(a) 7 (b) 8
(c) 9 (d) 14

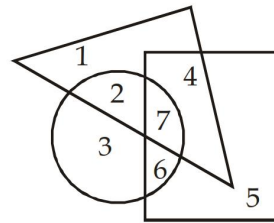
47. Who among the following is a male, urban oriented and also a civil servant but not educated ?

- (a) 13 (b) 12
(c) 6 (d) 10

48. Who among the following is a male civil servant, who is neither educated nor belongs to urban area?

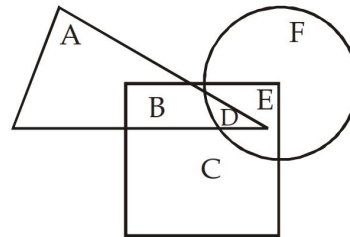
- (a) 7 (b) 13
(c) 4 (d) 1

49. If tall is equivalent to circle, Armymen to triangle and strong to square, indicate which number will represent strong Armymen ?



- (a) 3 (b) 4
(c) 5 (d) 6

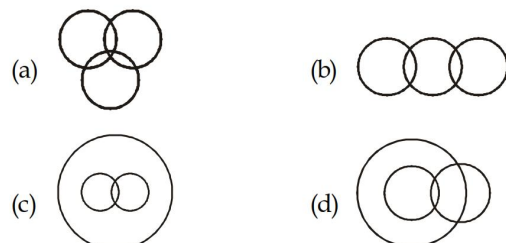
50. If the given figure, the triangle represents girls, the square represents sports persons and the circle represents coaches. The portion in the figure which represents girls who are sports person but not coach is the one labelled.



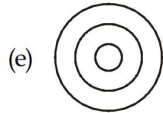
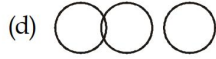
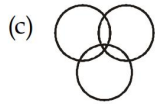
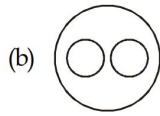
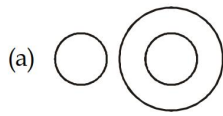
- (a) A (b) B
(c) D (d) E

Distinct Questions

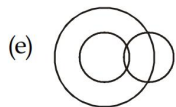
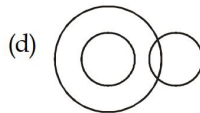
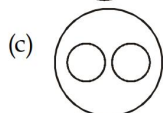
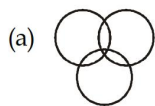
51. Which Venn diagram represents the inter-relationship among anti-social elements, Pickpocketers, Blackmailers ?



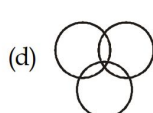
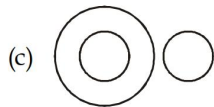
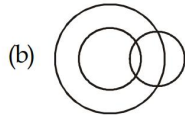
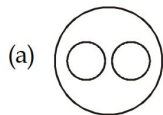
52. Which of the following diagram represents the relation among balls, footballs and spheres most appropriately ?



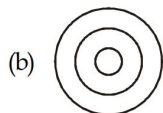
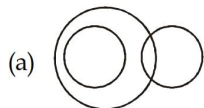
53. Which diagram represents the relationship between Women, Mothers and Engineers ?



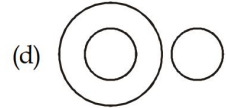
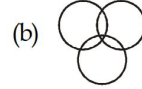
54. Which one of the following diagrams best depicts the relationship among cygnet, birds and rat?



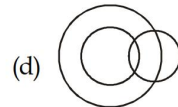
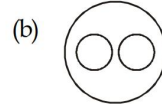
55. Which one of the following figures represents the relationship among Red, Clothes and Rose ?



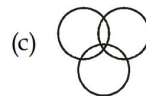
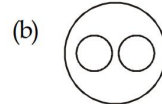
56. Which of the following diagram best depicts the relationship between Sire, Hawk and Boa?



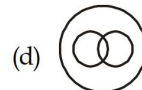
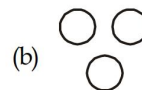
57. Which of the following diagrams represent the relationship among Steed, Adjutant and Aligator ?



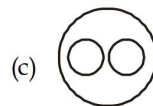
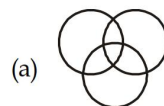
58. Which Venn diagram represents the inter-relationship among Spices, Cumin and Cloves ?



59. Which of the following diagram indicates the best relation between examination, question and practice?

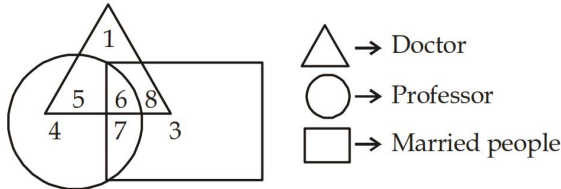


60. Which of the following diagram indicates the best relation between Teacher, Writer and Dancer.



Previous year Questions (Memory Based)

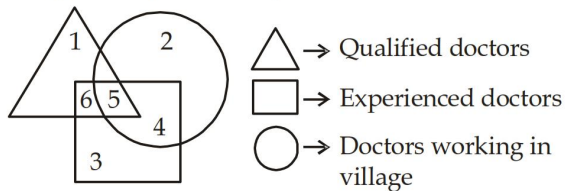
1. Which number indicates doctors who are not married.



- (a) 5 (b) 1
(c) 6 (d) 8

2. Study the diagram given below and answer the question.

The qualified and experienced doctor working in villages are represented by.



- (a) 4 (b) 5
(c) 2 (d) 6

3. The experienced doctors who are not working in villages.

- (a) 6 (b) 5
(c) 15 (d) 1

4. Only qualified doctors who are working in villages.

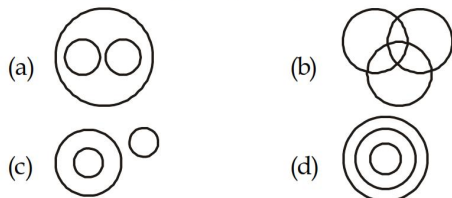
- (a) 2 (b) None
(c) 5 (d) 4

5. The doctors working in village are neither qualified nor experienced.

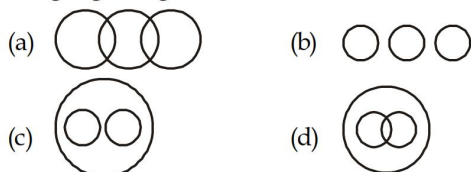
- (a) 15 (b) 10
(c) 2 (d) 4

(6-8) : Identify the diagram that best represents the relationship among classes given below.

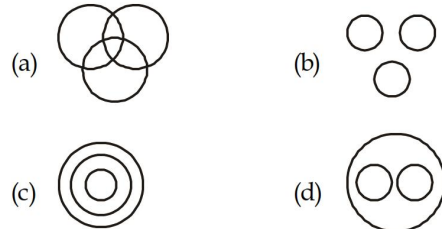
6. Professors, Researcher, Scientists.



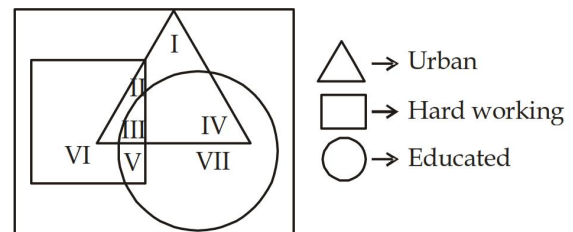
7. Language, English and Hindi.



8. Tigers, Lions, Animals.



9. Which one of the areas marked I-VII represents the urban educated who are not hard working?



- (a) I (b) III
(c) II (d) IV

10. All urban and hard working, educated shown by.

- (a) III (b) II
(c) IV (d) VII

11. Only hard working & only urban are shown by.

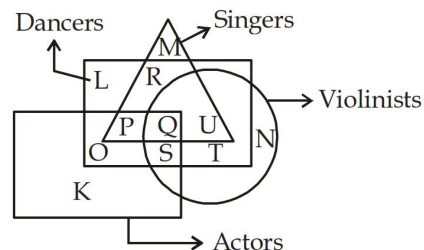
- (a) I, III (b) IV, III
(c) I, VI (d) II, III

12. Neither educated nor urban shown by.

- (a) VII (b) VI
(c) III (d) II

(13-14) : In the given figure, which letter represent

13. Those actors who are also dancers, singers as well as violinists.

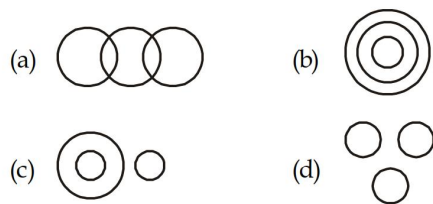


- (a) S (b) Q
(c) P (d) U

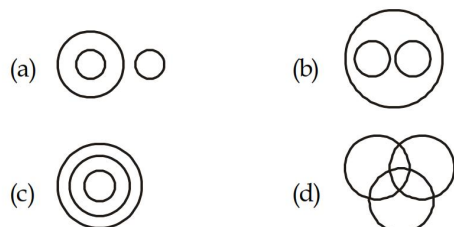
14. Those singers who are dancer but not (Actor and Violinists)

- (a) R (b) U
(c) T (d) K

Direction (15 – 27): Find out the diagram which indicate best relationship among given object.

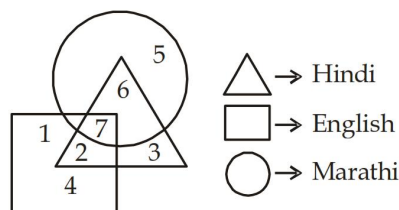


15. Travellers, Train, Bus.
16. Gases, Oxygen, Gold.
17. Indians, Bengali, East Indian.
18. World, India, Rajasthan.
19. Cricket, table tennis, Football.
20. Metal, Air, Road.
21. Nurse, Females, Doctor.
22. Shark, Bird, Lion.
23. Insect, Virus, Bacteria.
24. Milky-way, Earth, Universe
25. Cousins brothers female, Aunt.
26. Society, Family, Members.
27. Metal, Silver, Water.
28. Which diagram shows best relationship among carbon, sulphur dioxide and gases.



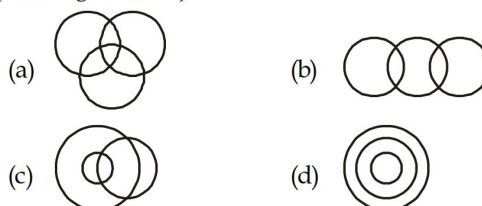
Direction (29 – 34): Given diagram shows different entities, find out relationship which are given below.

29. How many person speak minimum two languages.



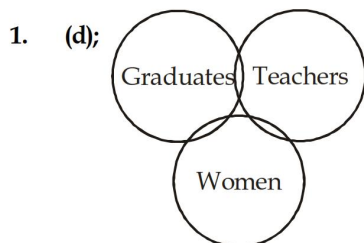
- (a) 8 (b) 15
(c) 4 (d) 11
30. How many persons speak all three languages.
(a) 17 (b) 9
(c) 11 (d) 7
31. How many persons doesn't speak marathi.
(a) 6 (b) 15
(c) 10 (d) 17
32. How many persons speak only 1 language.
(a) 12 (b) 13
(c) 15 (d) 20
33. How many persons speak hindi.
(a) 17 (b) 18
(c) 20 (d) 25
34. How many persons speak only marathi.
(a) 10 (b) 12
(c) 5 (d) 4

(35-40) : Find out best diagram which shows the relationship among these objects.

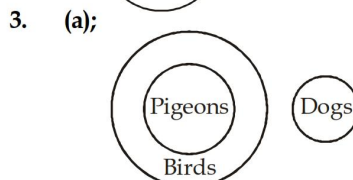
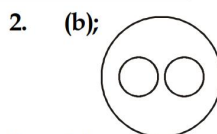


35. Mammal, Cat, Pet animal.
36. Male, Hindu, Muslim.
37. College, Lab, Equipment.
38. Letters, Word, Sentence
39. Men, female, professors.
40. Men, fathers, Indian.

Practice Set Solutions

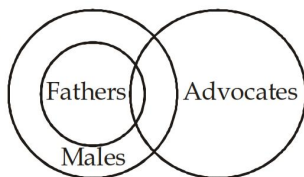


Here, some teachers may be women and some women may be graduates. Also some graduates may be teachers.



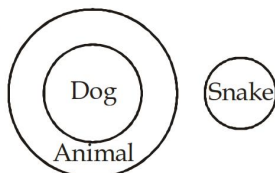
We can include pigeons into birds category but dogs are animals.

4. (a);



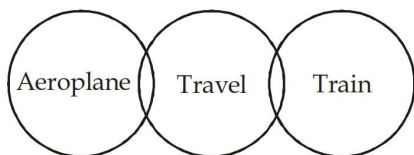
All fathers are males. Some males and some fathers may be advocates.

5. (a);



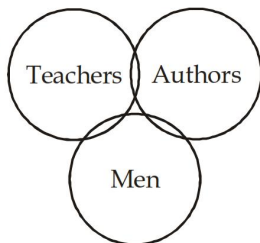
All dogs are animal but snake belongs to reptiles category.

6. (c);



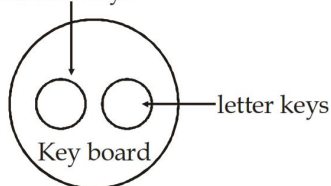
Travel can be both in Aeroplane and Train.

7. (b);



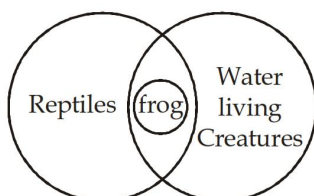
Some Authors may be men and some men may be teachers. Also, some teachers may be authors.

8. (b); function keys



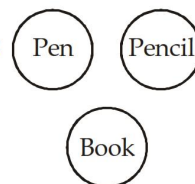
Function keys and letter keys both are present in key board but function keys are different from letter keys.

9. (a);



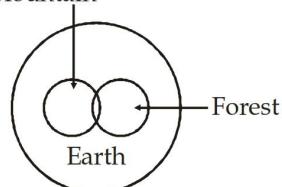
Some animals are belong to both categories reptiles and water living creatures. Such snake, crocodile, frog etc.

10. (b);



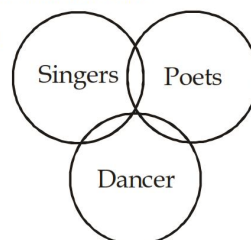
All are related to stationery but different from each other.

11. (c); Mountain



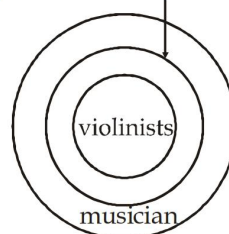
Mountain and forest both are on earth. But some forest may be on mountains.

12. (a);

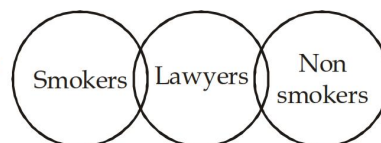


Some singers may be poets and some dancer may be singers. Some poets may be dancer.

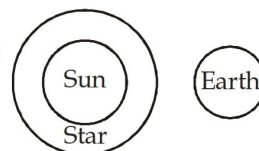
13. (b); instrumentalists



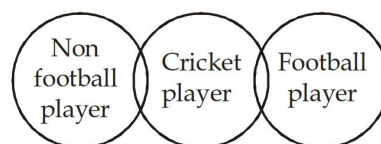
14. (a);

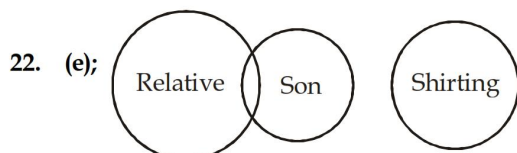
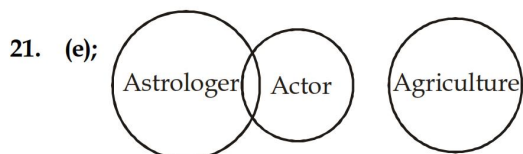
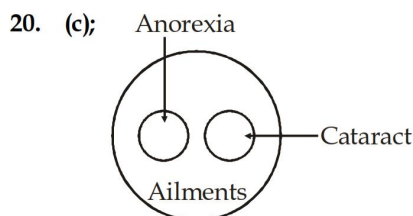
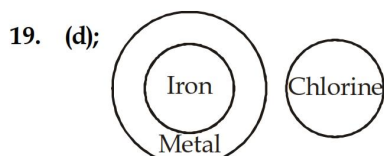
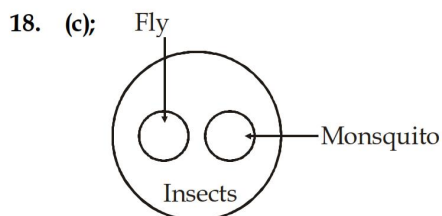
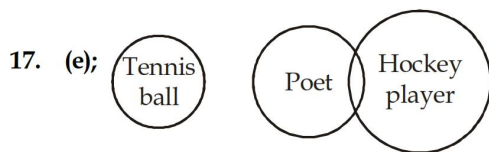


15. (d);



16. (a);

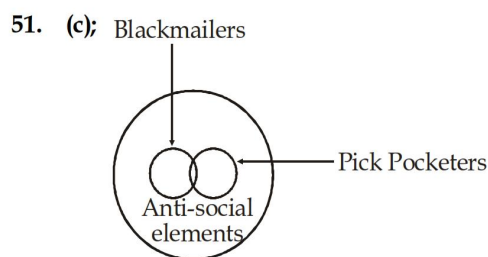




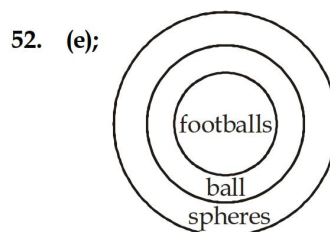
23. (c); Number 6 represents non-school going villagers infants.
 24. (a); Number 8 represents villagers boys.
 25. (b); Number 5 represents the school going villagers boys.
 26. (e); 4 represents the infant boys who are school going.
 27. (b); 3 represents the infant villagers boys who are school going.
 28. (d); Letter 'c' represents the blank space.
 29. (a); Letter 'e' represents the students who play volleyball and hockey.
 30. (d); No letter is there which represents the boys who play all games.
 31. (c); Letter 'f' represents the students who play football and hockey.

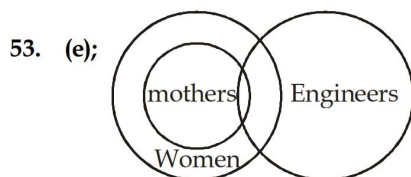
32. (d); None.
 33. (c); Letter 'f' represents the lawyer women.
 34. (b); Letter 'k' represents the men-doctor.
 35. (a); Letter 'h' and 'a' represents the women who are not lawyer and doctor.
 36. (c); Letter 'b' and 'c' represents the men who are not doctor and lawyer.
 37. (b); Letters 'd' and 'g' represents women doctors.
 38. (c); Tiger is a carnivorous animal and elephant is not carnivorous
 39. (d); Some students are boys and some are girls. So option d is the answer.
 40. (b); 7 Boys are disciplined students.
 41. (d); → 11.
 42. (b); → 3.
 43. (c); → 10.
 44. (a); → 4.
 45. (d); → 12 → The condition which is not mentioned should not be considered or assumed. For instance, here 6 also denotes the required region. But, since it lies inside the square and there is no mention of 'civil servant', so it cannot be the answer.
 46. (a); → 7.
 47. (c); → 6.
 48. (b); → 13.
 49. (b); → 4.
 50. (b); → B.

Distinct Solutions

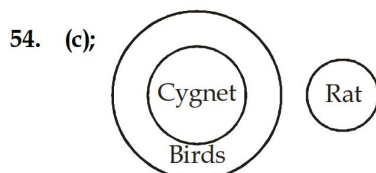


Some blackmailers may be pickpocketers but all blackmailers and all pickpockets are anti-social elements.

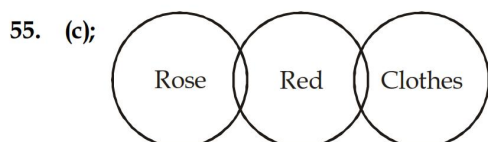




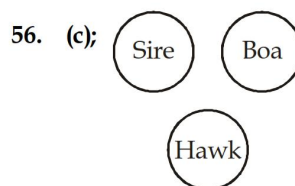
All mothers are women. Some mothers and some women may be Engineers.



Cygnets (the young of a Swan) are birds. And we can put it into birds category. But Rat does not belong to birds category.

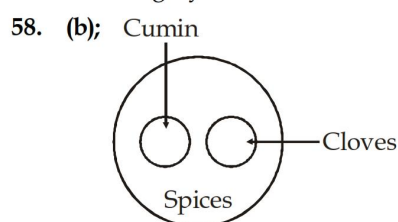


Some clothes and rose may be red. But rose can not be clothes.



All are related to different categories.

57. (c); All three belong to different categories. Alligator related to reptile category. Adjutant related to birds category and steed related to animal category.



Cumin and cloves both are related to spices category. But Cumin can not be cloves.

59. (c); The question is related to examination and practice both but partially.

60. (a); Some teachers are writers and they also are related to dancers.

Previous year Solutions (Memory Based)

1. (c); 6, the doctor who are not married (5 + 1).
2. (b); The qualified and experienced doctors working in villages are 5.
3. (a); The experienced doctor who doesn't work in villages are represented by 9.
4. (b); The only qualified doctors who are working in village is none.
5. (c); The doctor working in village who neither qualified nor experienced are represented by 2.
6. (b); Some professors are researchers as well as scientists so option (b) is correct.
7. (c); English and Hindi, both are languages, so answer is (c).
8. (d); Lions and tigers, both are animals, so answer is (d).
9. (d); The urban educated who are not hard working are shown by IV.
10. (a); The part III shows urban, hard working and educated.
11. (c); Only hard working with urban shown by I and VI.
12. (b); Neither educated nor urban are shown by VI.
13. (b); The actors who are also dancers, singers and violinists are shown by Q.

14. (a); R is a part of singers who are dancers but doesn't relate to actors and violinists.
15. (a); Some travellers are related to train and some are from bus.
16. (c); Oxygen is totally related to the gas and gold is a metal.
17. (b); Some Indians are Marathians and some are east Indians so answer is option (b).
18. (b); Rajasthan is a part of India and India is a part of world, so answer is (b).
19. (d); All are different.
20. (d); All three objects are unrelated to each other.
21. (a); Some female candidates are nurses and some are doctors.
22. (d); Shark, bird and lion, all of them belong to different areas.
23. (d); Insect, virus and bacteria all are different, they don't have any common relation.
24. (b); Our earth is present in the Milky Way galaxy and the Milky Way is a part of the universe.
25. (c); All aunts are female and cousin brother is different from both of them.
26. (b); Members are part of family, and family is a part of society.

27. (c); Silver is a metal and water is different from them.
28. (b); Sulphur dioxide and carbon both are part of gases.
29. (b); The person who speak min two language are $2 + 7 + 6 = 15$.
30. (d); 7 person can speak all three language.
31. (c); The person who doesn't speak marathi are $4 + 2 + 3 + 1 = 10$
32. (b); The person who speak only one language = $4 + 3 + 1 + 5 = 13$.
33. (b); Total person who speaks Hindi are $6 + 7 + 5 = 18$
34. (c); The person who speak only marathi is 5.
35. (c); All cats are mammal but some cat and mammals are pet animals.
36. (b); Some male candidates are hindu and some are Muslim.
37. (d); Equipment is a part of lab and lab is present in college.
38. (d); Combination of letters is called word, and combination of words is called sentence.
39. (b); Some professor are male and some are female.
40. (c); All father are men and some men are Indians.